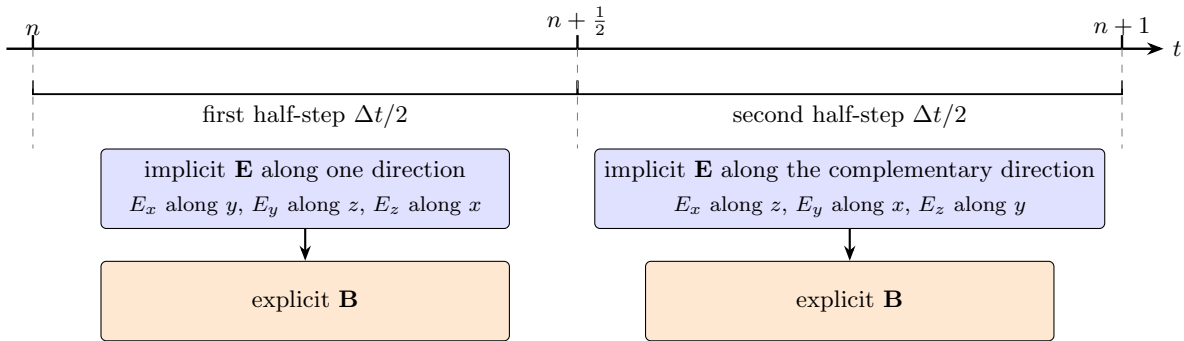
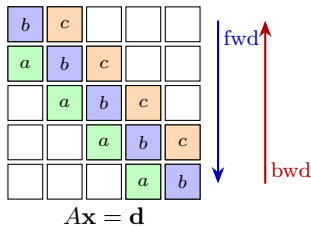




each implicit E line solve is $a_i x_{i-1} + b_i x_i + c_i x_{i+1} = d_i$



Thomas algorithm ($O(n)$)

1. Forward sweep

for $i = 2, 3, \dots, n$:

$$w := a_i/b_{i-1}, \quad b_i := b_i - wc_{i-1}, \quad d_i := d_i - wd_{i-1}$$

2. Back substitution

for $i = n - 1, n - 2, \dots, 1$

$$x_n = d_n/b_n, \quad x_i = (d_i - c_i x_{i+1})/b_i$$